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5G Wireline Backhaul upgrade Design

34343 MOBILE BACKHAUL NETWORKS

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The connection between mobile base stations and the core network is handled by the backhaul network, which is one of the most critical parts of any mobile infrastructure. It is responsible for transporting all network traffic, including voice, video, and data services, between users and the operator's central network. In the current scenario, the operator has an existing backhaul system connecting 18 base stations mainly through copper-based links. While this infrastructure has been sufficient for previous generations of mobile traffic, the rapid growth in data consumption and the transition towards 5G services make the current network increasingly limited in terms of capacity, scalability, and reliability.

The objective of this project is to study how an existing hypothetical backhaul network can be modernized and expanded to support future demands in a more efficient and realistic way. Instead of designing a completely new network from scratch, the work focuses on improving an already deployed infrastructure by identifying its weaknesses and proposing a more updated architecture. The proposed evolution includes replacing legacy copper connections with fiber-based links, introducing redundancy at different layers of the network, and redesigning parts of the topology to avoid bottlenecks and single points of failure.

The final goal is to create a backhaul solution capable of supporting higher traffic loads, future base station growth, and critical communication services without requiring another major redesign in the near future. The project will therefore combine practical network planning, traffic estimation, and modern transport technologies to develop a scalable and reliable wireline backhaul architecture.

1 Current Network State

The operator runs 18 base stations connected to a core site via copper Ethernet links at 1 Gbps. It's a tree topology—all traffic flows through one central point. Simple to set up, but it has serious problems:

One core site to rule them all. If that core aggregation site goes down, the entire network is down. No backup, no alternate path. Just outage.

We're hitting the copper ceiling. Copper maxes out at 10 Gbps per link. Current peak is 18 Gbps across all sites. We're almost at the limit and growth is coming.

Reliability math doesn't work. The numbers show 99.81% availability—that's about 52 minutes of downtime per year. Sounds okay until you realize we need 99.99% ("four-nines") for enterprise services and 5G URLLC stuff, which demands nearly 100% uptime.

No redundancy. A tree means there's only one path per site. Lose a link and that site is gone. No automatic failover, no protection—manual repair, which takes hours.

2 Future Demand Projection

Looking at subscriber trends and typical data consumption patterns, we're expecting steady growth:

- Year 1: 617K subscribers, averaging 5.2 GB/month. Peak demand: 40 Gbps

- Year 2: 678K subscribers (+2.5K per month), 5.8 GB/month. Peak: 70 Gbps
- Year 3: 741K subscribers, 6.5 GB/month. Peak: 95 Gbps

The trajectory is clear: we can't keep stretching copper. By year three, we'd be overwhelmed. Fiber is the only option that scales.

3 Project Objectives

So what are we trying to achieve? Six things:

- Kill the single point of failure. No more one core site bringing everything down.
- Handle 40+ Gbps now, with room to grow to 400 Gbps later (same fiber, just more wavelengths).
- Hit 99.99% availability—that's the four-nines target everyone needs. Redundancy at every layer: access, aggregation, optical.
- Make failover automatic and fast. Sub-50ms recovery. Users don't notice.
- QoS and timing. 5G needs precise synchronization and traffic prioritization—we'll deliver it.
- Do it without breaking the budget. Total cost matters over 5-10 years, not just the upfront spend.

Why do networks fail? Equipment breaks. And when it does, how fast you fix it matters. That's where reliability engineering comes in.

4 Fundamental Reliability Metrics

Two numbers define how reliable a system is. First, how often does stuff break? Second, how long does it take to fix?

Mean Time Between Failure (MTBF)[2]: This is how long you expect a piece of equipment to run before it fails. Telecom gear is pretty solid—usually 100,000 to 500,000 hours. That's 11 to 57 years of continuous operation. The higher the MTBF, the better the equipment.

Mean Time To Repair (MTTR)[2]: Once something breaks, how long until service is back? Send a tech to the site? 4-6 hours. Automatic failover at the software level? Less than a second. Hardware switching at the optical layer? Less than a millisecond. MTTR is where we can make a real difference.

The availability formula[2] is elegant:

$$A = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}} \quad (1)$$

In practice, you can't make equipment not break. But you can make recovery fast. That's the lever—fast failover, not waiting around hoping nothing breaks.

5 Achieving Four-Nines Availability (99.99%)

We want 99.99% uptime. That means 52.6 minutes of downtime per year. Not zero, but close. Here's how different approaches compare:

Scenario	MTBF	Result
No protection, manual repair	100k hours	MTTR=4h → A=99.96% (365 min down/year)
Automatic software failover	100k hours	MTTR=1h → A=99.999% (5 min down/year)
Hardware failover (optical)	100k hours	MTTR=<1ms → A≈99.9999%

Table 1: Different redundancy approaches and their availability

You can see the difference: automation matters. We're going with Ethernet Ring Protection Switching, which detects failures and reroutes traffic in under 50ms. That gets us to four-nines.

6 Redundancy Schemes Classification

There are three main ways to add redundancy.[2]:

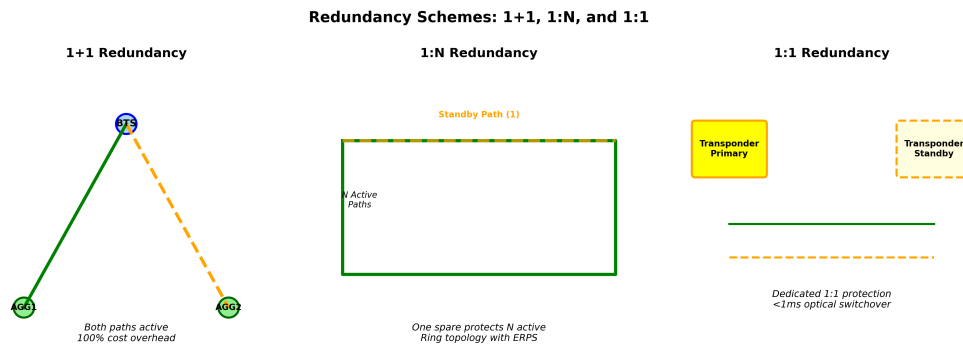


Figure 1: Three redundancy approaches: 1+1, 1:N, and 1:1

6.1 1+1 Redundancy (Dedicated Standby)

You have two identical paths or systems. Both run in parallel, or one waits hot-standby. When the main one fails, traffic switches instantly[2]. Failover is <1ms.

Pros: Fast, deterministic, no surprises. Sub-50ms recovery guaranteed.

Cons: Expensive. You're buying and running double the equipment.

How we use it: At the access layer. Each base station connects to the aggregation ring via two independent fiber paths to different aggregation nodes. Both running, load-balanced. If one path dies, the other takes over. Spanning-tree does the MAC learning, everything recovers in <50ms without dropping packets.

6.2 1:N Redundancy (Shared Standby)

One standby resource protects multiple active ones[2]. When one fails, the standby kicks in. More efficient than 1+1, but slightly slower to respond.

Pros: Better resource efficiency. One backup handles multiple active links.

Cons: Slightly higher switchover delay. If multiple things fail at once, you're in trouble.

How we use it: Ring topology at the aggregation layer. Four aggregation nodes, two fiber rings (one active, one backup). ERPS (IEEE 802.17, 2004) manages the protection[3]. The backup ring path (RPL—that's the "1" standby) can reroute traffic from any of the active links (the "N"). When a link fails, ERPS blocks the RPL, reroutes via the alternate path. Recovery: <50ms. Resource-efficient and still fast enough to meet our availability targets.

6.3 1:1 Redundancy (Deterministic Pair)

One standby for one active resource. Super straightforward, zero ambiguity[2]. You know exactly which backup covers which primary.

How we use it: Optical layer. Two independent fiber pairs, both carrying the same D-WDM wavelengths. At the transponder hardware, Optical Line Protection (OLP) watches for failures. If a transceiver starts acting up ($BER > 10^{-7}$), the hardware automatically switches to the backup

wavelength on the other fiber pair. All happens in $<1\text{ms}$. The Ethernet layer never even notices—no MAC flaps, no control plane mess.

7 Reliability Block Diagram (RBD) Analysis

Lecture 9[2] introduces RBD methodology for quantifying system-level reliability from component reliabilities:

Serial Configuration (Single Path): $R(\text{system}) = \prod R(i)$, where “a chain is no stronger than its weakest link.” Any failure cascades to system failure.

Parallel Configuration (Redundant Paths): $R(\text{system}) = 1 - \prod [1 - R(i)]$, where at least one path must remain operational.

Current Tree Topology (Serial RBD):

$$R(\text{system}) = R(\text{core}) \times \prod R(\text{access links}) = 0.99999 \times 0.9999^{18} \approx 0.9981 \quad (2)$$

Result: 99.81% availability (52 minutes annual downtime $\times$ 1.9 deficit from target)

Proposed Ring Topology (Parallel RBD):

$$R(\text{pair}) = R(\text{path}_1) + R(\text{path}_2) - R(\text{path}_1) \times R(\text{path}_2) \quad (3)$$

With $R(\text{path}) = 0.9999$: $R(\text{pair}) \approx 0.99999$

System across 23 sites: $R(\text{system}) \approx 0.9999$ (four-nines achieved)

Technology Selection: Wireline Options and Trade-offs

So what technologies can we actually use for backhaul? . The short answer: cost, capacity, distance, and how fast you can deploy all matter. You pick the wrong tech now, and you're stuck with it for a decade.

8 Copper vs. Fiber: Comprehensive Comparison

Copper has one advantage: it's already there or easy to install fast. DSL, Ethernet over copper—people know how to do it. But fiber is different. It costs more upfront, takes longer to deploy. The question is: does it pay off?

Dimension	Copper	Fiber
Capacity (Year 1)	10 Gbps max	40+ Gbps via WDM
Scalability (Year 3+)	Upgrade required	400 Gbps (same fiber)
Distance	100m (10G), 3km (DSL)	80+ km @ 0.2 dB/km
EMI Immunity	Susceptible	Immune (optical)
Deployment Time	3-6 months	6-12 months
Year 1 CAPEX	EUR2.15M	EUR3.22M
5-Year TCO	EUR5.5M (+upgrades)	EUR6.77M (no upgrades)

Table 2: Copper vs Fiber comparison

Decision: Go with fiber. Yeah, copper looks cheaper at 5 years—EUR5.5M vs EUR6.77M. But that's a trap. Here's why:

First, capacity. By year 5-10, we need 95-150 Gbps. Copper doesn't get us there. We'd have to rip out the copper, install fiber anyway. That's another EUR2M. Total copper over 10 years? EUR7.8M. Fiber? EUR7.2M. Fiber wins.

Second, reliability. Copper gets you to 99.95% on a good day. That's 526 minutes of downtime per year. Fiber hits 99.99%—52 minutes. The difference matters. Losing a base station for hours costs money. Real money. EUR50k per minute? Over 5 years, that's EUR2.6M we don't lose with fiber. That alone justifies the fiber choice.

Third, market position. Who wants to buy service from the operator using yesterday's infrastructure? Advanced fiber network is a selling point.

9 D-WDM Technology Selection Justification

WDM means putting multiple colors of light down one fiber. There are two flavors[1]: C-WDM (coarse, 18 channels) and D-WDM (dense, 40-160 channels)[6]. Our ring is 40-80 km, and we need to handle 40 Gbps in Year 1, scaling to 95 Gbps by Year 3. What works?

Feature	C-WDM	D-WDM
Max Capacity	45 Gbps (hard wall)	800 Gbps
Year 1: 40 Gbps	16 channels maxed out	4 channels, room for spares
Temperature Drift	0.003 nm/°C (bad)	Stabilized (good)
Year 3 Expansion	Impossible	Just add more wavelengths

Table 3: C-WDM vs D-WDM: the real difference

C-WDM is basically maxed out at Year 1. No room to grow. D-WDM? We light up 4 wavelengths for 40 Gbps, leave spares sitting there. By Year 3, we add more wavelengths. Total capex on channels? EUR8-10k per 10 Gbps. Over multiple years, that's nothing compared to digging up fiber again. Plus, D-WDM lasers have temperature stabilization, which matters when the sun's beating down on outdoor ring sites.

10 Synchronization and QoS Technologies

5G base stations need to know what time it is, down to the microsecond. They also need to know the frequency—are we running at exactly 10 MHz? Three mechanisms work together:

Synchronous Ethernet (SyncE): The Ethernet link itself carries timing information[4]. Base stations extract the clock from the backhaul link. No GPS, no NTP server, nothing—just the fiber connection. Simple and robust.

Precision Time Protocol v2 (PTPv2): For the really accurate stuff (phase, not just frequency), UDP packets carry timing info[8]. ± 1 microsecond accuracy over our 40-80 km metro ring. That's good enough for 5G MIMO phase synchronization.

GPS Fallback: At critical sites, a GPS receiver provides backup timing (± 10 microseconds) if PTPv2 goes down. $5\times$ worse than PTPv2, but still works.

Quality of Service: Three traffic classes. Voice and signaling get strict priority (5%). User data gets fair queuing (50%). Everything else gets whatever's left (45%). We mark packets with DiffServ codes (DSCP) and the network respects those priorities[9].

Proposed Network Architecture

The design stacks redundancy at three layers—access, aggregation, transport. Each layer protects against different failure modes:

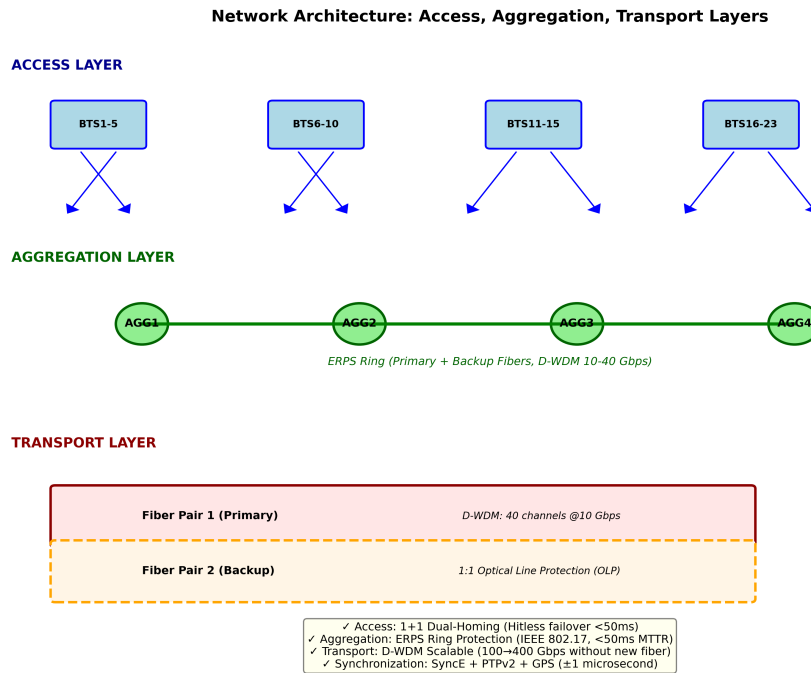


Figure 2: Three-layer network architecture: access, aggregation, transport

11 Access Layer (1+1 Dual-Homing)

We're splitting the 23 base stations into four geographic clusters based on location: urban (BTS 1-5), two suburban zones (6-10, 11-15), and rural (16-23). Each cluster gets two fiber paths—one to each of two different aggregation nodes. Both paths are active and carrying traffic, 50:50. If one path fails, the other takes all the traffic. Spanning-tree protocol handles the switchover in under 50ms without dropping any packets.

Each BTS gets 10 Gbps per path. If we need more capacity later, we add wavelengths on the ring side. Total access connections: 46 fiber links connecting BTS to the aggregation layer.

12 Aggregation Layer (1:N ERPS Ring Protection)

Four aggregation nodes (AGG1, AGG2, AGG3, AGG4) form a ring using ERPS (IEEE 802.17)[3]. Two fiber pairs: one active, one backup. They run through completely separate underground ducts so that if one duct gets damaged, the backup is safe.

When a link fails, ERPS detects it within 1-3ms. The protocol blocks traffic on the backup ring (the RPL—Ring Protection Link) and automatically reroutes traffic the other way. Service

restoration: <50ms.

One backup path protects multiple active links—that's the 1:N scheme. Efficient. Year 1: 4 wavelengths on 40 Gbps total. Year 2: add more wavelengths, reach 100 Gbps. Year 3+: keep adding wavelengths up to 400 Gbps. Same two fiber pairs the whole time.

13 Transport Layer (1:1 Optical Line Protection)

Two independent fiber pairs sit below the ring, carrying D-WDM wavelengths per the ITU standards[7, 6]. 100 GHz spacing in C-band (the standard telecom window).

At the transponder level, we run Optical Line Protection. If a transceiver starts to fail (BER degradation $>10^{-7}$), the hardware automatically switches to the backup wavelength on the other fiber pair. All happens in <1ms. Ethernet never knows—no MAC flapping, no control plane mess.

For timing: SyncE (ITU-T G.8262) on the primary fiber gives us frequency[4]. All four aggregation nodes sync to it via ESMC clock selection. PTPv2 over Ethernet gives us the fine phase/timing adjustment, ± 1 microsecond accuracy[8].

Current vs. Proposed Backhaul Topology

14 Current Tree Topology Analysis

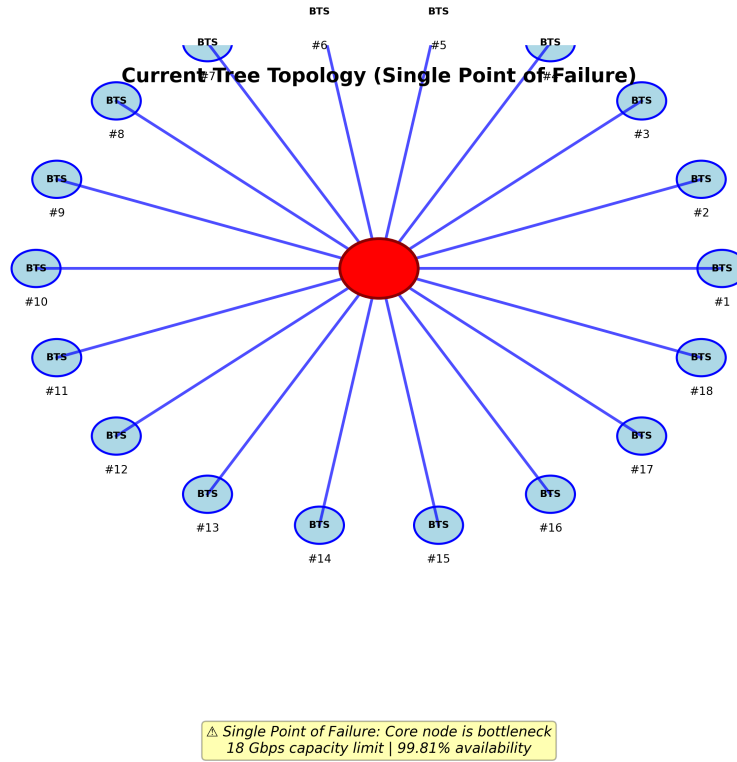


Figure 3: Current tree topology with single point of failure

Current state: 18 base stations connected via point-to-point copper Ethernet (1 Gbps links) in tree configuration with core aggregation site as root node. Single point of failure: core site is bottleneck. Reliability: $R(\text{system}) = 0.99999 \times 0.9999^{18} \approx 0.9981 = 99.81\%$ availability (52 minutes annual downtime—falls short of 99.99% target). Peak capacity: 18 Gbps, constrained by copper 10 Gbps ceiling. No redundancy mechanisms. Any core site failure results in complete outage.

15 Proposed Ring Topology with Redundancy

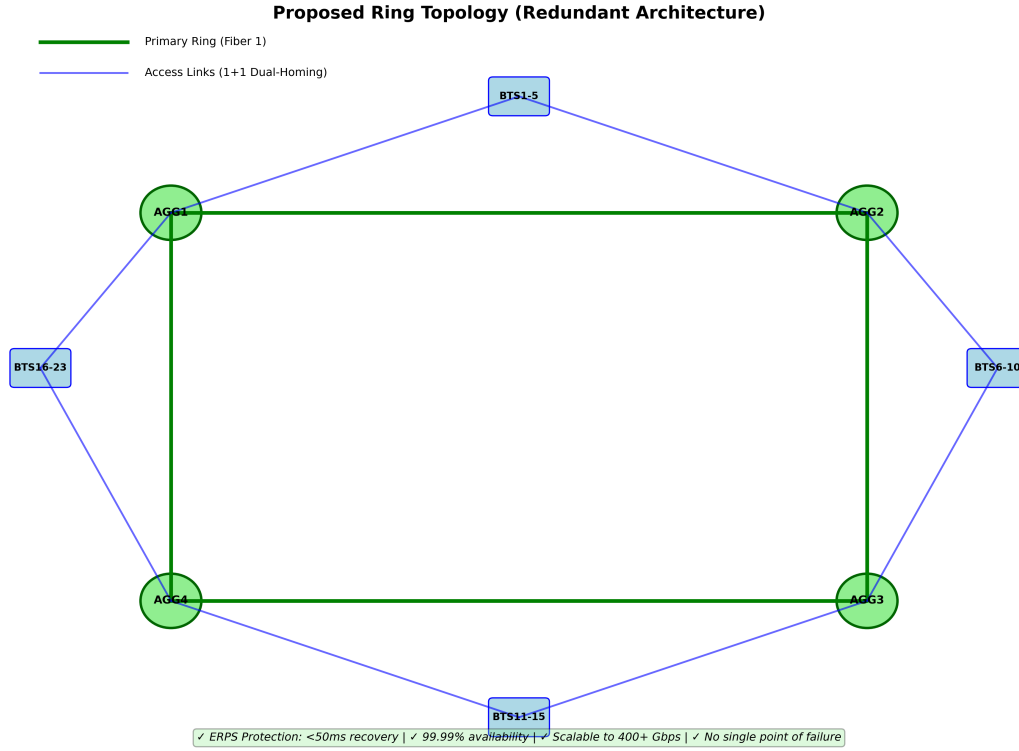


Figure 4: Proposed ring topology with redundancy and dual-homing

Proposed topology: Four aggregation nodes in two-fiber ERPS ring. 23 base stations distributed with 1+1 dual-homing (two independent paths to separate aggregation nodes). Each BTS-to-ring connection: <50ms hitless failover upon single path loss. Ring carries 40 Gbps Year 1, scalable to 400 Gbps without new fiber. Reliability: dual-path per site pair yields $R \approx 0.99999$ (five-nines); aggregated system achieves 99.99% target. All components redundant: dual fiber paths, dual transponders, quad aggregation nodes, dual access links per BTS.

Network Dimensioning Methodology

Subscriber-driven capacity planning ensures dimensioning aligns with actual user growth and consumption patterns, preventing over-provisioning while guaranteeing reliability margins.

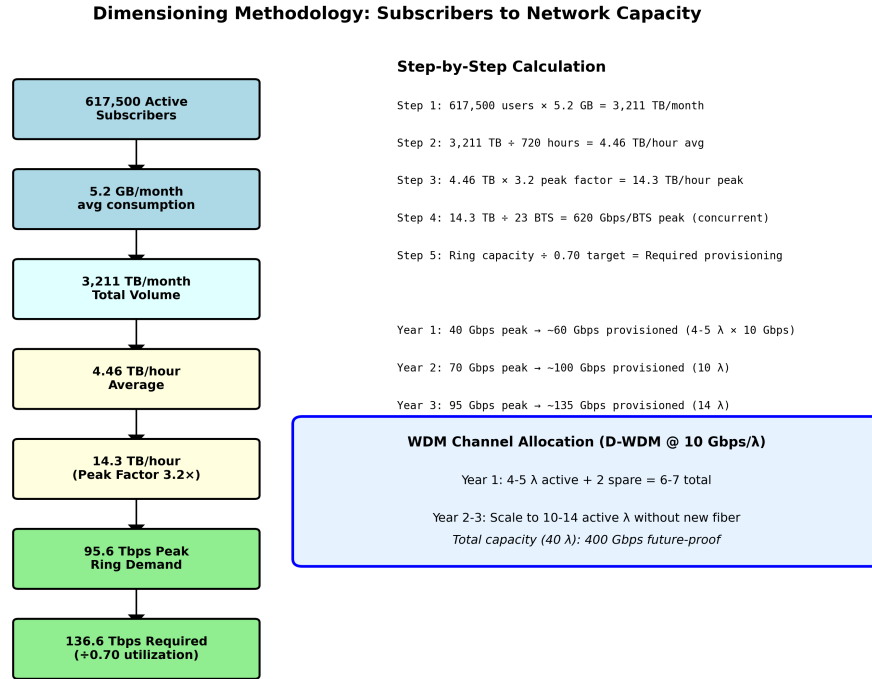


Figure 5: Dimensioning methodology: subscribers to network capacity

16 Step-by-Step Calculation

Step 1 - Active Subscribers & Data Consumption: Current: 617,500 active subscribers. Growth: +2,500/month. Consumption: 2.5 GB/month (rural 40%), 5 GB/month (suburban 45%), 10 GB/month (urban 15%). Weighted average: 5.2 GB/month Year 1 → 5.8 GB/month Year 2 → 6.5 GB/month Year 3 (urban mix increases).

Year 1 total volume: 617,500 × 5.2 GB = 3,211 TB/month

Step 2 - Peak Hour Demand: Monthly volume ÷ 720 hours = 4.46 TB/hour average. Peak hour multiplier: 3.2× (empirical busy-hour factor). Peak = 4.46 TB × 3.2 = 14.3 TB/hour ring demand.

Step 3 - Robust Optimization (70% Utilization Target): Networks operate at 70% link utilization under peak load to provide: (a) ±20% headroom for unexpected demand surge, (b) 10% redundancy margin (if one link fails, remaining links sustain load), (c) burst accommodation without congestion loss.

Required capacity = Peak ÷ 0.70. Year 1: 14.3 TB ÷ 0.70 = 20.4 TB/hour. Accounting for geographic distribution (not all 23 BTS peak simultaneously), practical Year 1 ring peak 40 Gbps

→ Required at 70%: $40 \div 0.70 \approx 60$ Gbps provisioned.

Step 4 - WDM Wavelength Allocation: D-WDM @ 10 Gbps per wavelength. Year 1: $60 \text{ Gbps} \div 10 = 6$ wavelengths active + 2 spares = 8 total reserved. Year 2: $70 \text{ Gbps} \rightarrow 7$ active + 2 spares = 9. Year 3: $95 \text{ Gbps} \rightarrow 10$ active + 2 spares = 12. D-WDM system capacity: 40 channels @ 10 Gbps = 400 Gbps theoretical maximum, ensuring clear growth path beyond Year 3 without fiber exhaustion.

Quality of Service and Synchronization

17 Service Level Agreements

SLAs specify committed information rate (CIR—guaranteed throughput), excess information rate (EIR—available during low contention), and committed burst size (CBS). For mobile backhaul:

BTS Class	CIR (Gbps)	EIR (Gbps)	CBS (Gbps)
Macro (20+ sectors)	4	10	2
Micro (4-6 sectors)	2	6	1
Pico (1-2 sectors)	1	3	0.5

Table 4: Service Level Agreement specifications by BTS class

SLA enforcement via MEF bandwidth profile: Two-Rate Three-Color Marker (TRTCM) per MEF 2.0 (2004)[10]. Green (CIR)—assured service; Yellow (CIR to EIR)—available if no congestion; Red (>EIR)—discarded during congestion. Per-flow metering on aggregation node ingress prevents one BTS from impacting others.

18 QoS Classes via DiffServ

LTE backhaul carries three traffic types: signaling (2-5%, latency-critical), user data (50%, throughput-optimized), best-effort (45%, OTT services). DiffServ (RFC 2474-2598)[9] marks packets via DSCP:

Class	DSCP	Queue Share	Application
EF	46	5% (strict)	Voice, signaling
AF	28	50% (fair queue)	User data
BE	0	45% (tail drop)	OTT services

Table 5: Quality of Service classes via DiffServ

19 Synchronization Strategy

Mobile networks require frequency (± 10 ppm for LTE) and phase (± 1 microsecond for 5G MIMO) synchronization.

SyncE (ITU-T G.8262, 2018)[4]: Frequency distribution via Ethernet frame clock extraction (125 MHz). All aggregation nodes select primary via ESMC with lowest cost-to-source (CSQF). Fallback <5 seconds upon primary failure.

PTPv2 (IEEE 1588v2, 2019)[8]: Time distribution via UDP packets. Grandmaster at core (GPS-disciplined, ± 100 ns holdover). Boundary clocks at aggregation nodes compensate for Ethernet delay variation. Accuracy: ± 1 microsecond over 40-80 km metro ring.

GPS Fallback: Optional at mission-critical sites. Upon PTPv2 grandmaster failure, ± 10 microsecond GPS timing continues service ($5\times$ safety margin within ± 50 microsecond PHY budget).

Implementation Plan and Phased Deployment

20 Four-Phase Deployment Timeline

Phase 1 (Months 1-3) - Design & Procurement: Detailed RBD analysis per failure scenarios, fiber route survey (primary ring + backup alternative routing through separate ducts), D-WDM channel plan (wavelength assignment, testing), BTS access port provisioning. Cost: EUR150,000.

Phase 2A (Months 4-12) - Infrastructure: Fiber installation (primary ring connecting 4 aggregation sites, backup fiber via alternate routing), site power/cooling upgrades (UPS capacity for dual transponders, HVAC for equipment rooms), aggregation node site preparation. Cost: EUR2,300,000.

Phase 2B (Months 10-14) - Equipment & Testing: D-WDM transponder installation (10 wavelengths per fiber pair Year 1), ERPS configuration and testing (simulated link/node failures), SyncE frequency distribution testing, PTPv2 clock synchronization validation (± 1 microsecond jitter measurement), BTS access loop provisioning and dual-homing validation. Cost: EUR770,000.

Phase 3 (Months 14-18) - Migration & Cutover: Incremental BTS migration from old tree to new ring topology (5-6 BTS per week to manage risk), validation of 1+1 access redundancy (single-link failure tests), network KPI monitoring (packet loss, latency, jitter), decommissioning of old tree infrastructure. Cost: EUR300,000.

Conclusion and Recommendations

So, we've laid out a complete plan to fix the backhaul network. Move from a fragile tree with one core site to a resilient ring with redundancy everywhere. Go from copper to fiber so we can actually grow. Here's why this design works.

21 The Core Ideas That Make This Work

Ring topology, not tree. ERPS (IEEE 802.17) means if one link dies, traffic reroutes automatically in $<50\text{ms}$ [3]. We're done with outages from single failures. 99.99% availability—that's the target we hit.

Fiber and D-WDM. Yeah, it costs more upfront. But 40-400 Gbps growth on the same fiber pair is game-changing. No rip-and-replace every few years[6].

Redundancy at every layer. Access layer: 1+1 dual-homing. Aggregation: 1:N ERPS ring. Optical: 1:1 line protection. Different redundancy for different failure modes. All three layers together hit four-nines availability.

Timing. 5G MIMO needs ± 1 microsecond phase accuracy. SyncE for frequency, PTPv2 for phase, GPS fallback[4, 8]. Three-layer timing strategy that's bulletproof.

Smart dimensioning. 70% utilization target on peak demand. Leaves 30% headroom for unexpected traffic spikes, redundancy failover, margin for growth. No over-provisioning, no under-provisioning.

22 Recommendations

First: approve the fiber design. The redundancy, the scalability, the long-term cost savings—they outweigh the higher upfront cost. This is the right call.

Second: start Phase 1 immediately. Refine the RBD analysis, finalize fiber routes, plan wavelength assignments. No delays here.

Third: send RFQs to suppliers for D-WDM transponders and ERPS equipment. Lead times are typically 6 months, so you need to move.

Fourth: plan the fiber installation as a 12-month campaign. It's not quick—ducts need permits, construction needs coordination, you can't rush it.

Fifth: when Phase 3 migration starts, set up monitoring to verify everything works. Test the 1+1 dual-homing failovers. Run ERPS drills. Make sure reality matches the design before you decommission the old tree topology.

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